

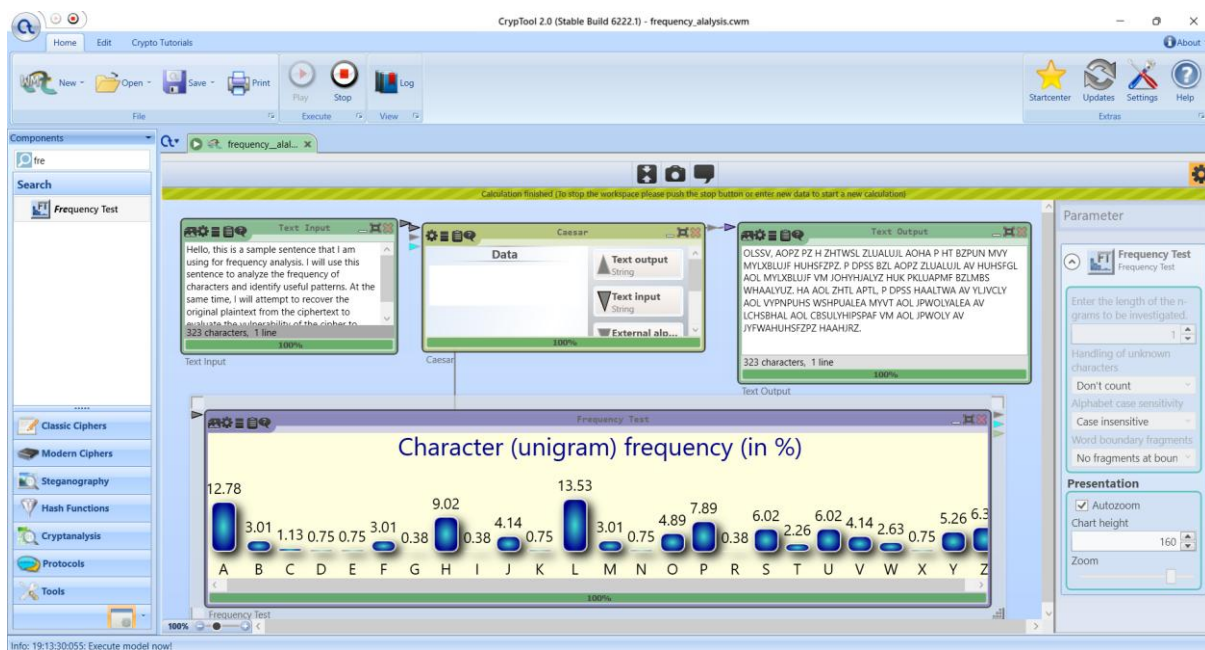
Frequency Analysis and Cryptanalysis of Caesar Cipher

Plaintext Used: Hello, this is a sample sentence that I am using for frequency analysis. I will use this sentence to analyze the frequency of characters and identify useful patterns. At the same time, I will attempt to recover the original plaintext from the ciphertext to evaluate the vulnerability of the cipher to cryptanalysis attacks.

1. Frequency Analysis

The plaintext was encrypted using the Caesar cipher and the resulting ciphertext was analyzed using the Frequency Test component in CrypTool 2. The frequency graph shows how often each character occurs in the ciphertext. Since the Caesar cipher applies the same fixed shift to every letter, the frequency pattern of the letters is preserved, although the letters themselves are shifted.

In the frequency distribution, L is one of the most frequently occurring characters in the ciphertext. In normal English text, E is one of the most frequently occurring letters. Therefore, we can hypothesize that the ciphertext character L corresponds to the plaintext character E.

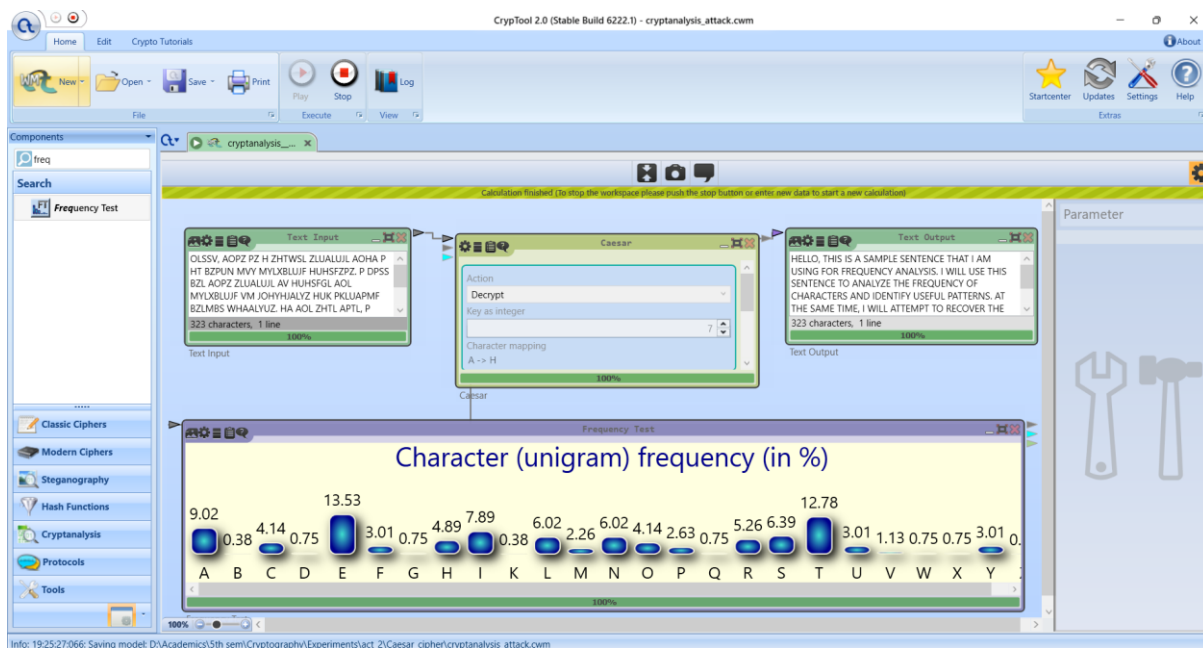


CrypTool 2 Frequency Analysis of Caesar Cipher

2. Cryptanalysis Attack

To determine the possible Caesar shift, the distance between the assumed plaintext character E and the ciphertext character L is calculated. The difference between E and L is 7 positions in the alphabet. Therefore, a shift of 7 is taken as the likely encryption key.

The ciphertext was then decrypted using a Caesar key of 7 in CrypTool 2. The resulting output produced a meaningful and readable English sentence, which matches the original plaintext. This demonstrates that frequency analysis can be used to recover the key and plaintext of a Caesar cipher.



CrypTool 2 Caesar Cryptanalysis with Key = 7

3. Observation

- E is one of the most frequently occurring letters in typical English text.
- L is one of the most frequently occurring characters in the given Caesar ciphertext.
- The distance between E and L is 7 positions, suggesting a Caesar shift of 7.
- Decrypting the ciphertext with key 7 produces the original readable plaintext.
- Therefore, the Caesar cipher is highly vulnerable to frequency-analysis-based cryptanalysis.

4. Conclusion

The experiment demonstrates that the Caesar cipher is vulnerable to frequency analysis because it preserves the relative frequency pattern of the plaintext letters. By comparing common ciphertext characters with common English letters and testing the resulting shift, an attacker can recover the encryption key and decrypt the message.